

# Kovid Sharma

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## EDUCATION

### JMI, DELHI

B.TECH IN COMPUTER  
SCIENCE

2020 -- 2024 | New Delhi  
CGPA: 9.27/10

### MODERN DPS

SENIOR SECONDARY  
2018 - 2020 | Faridabad  
92.6%

## COURSEWORK

- System Design & Architecture
- Data Structures and Algorithms
- Distributed Systems
- Database Management
- Object Oriented Programming
- Cloud Computing

## ACHIEVEMENTS

- **Knight** on LeetCode – peak rating 1855  
leetcode.com/kovid1010
- **World Rank 954**, Google HashCode
- **Grand Finalist**, Smart India Hackathon

## SKILLS

### Languages:

Java (8/17) • Python •  
C/C++ • SQL •  
JavaScript/TypeScript • Dart

### Backend & Design:

Spring Boot • Spring Cloud •  
Microservices • System  
Design • REST/gRPC APIs •  
Hibernate/JPA • Apache  
Kafka • Redis Caching •  
Concurrency

### Platform & Tooling:

Docker • Kubernetes • GCP  
(Cloud Run, Cloud SQL) •  
CI/CD (Jenkins, Cloud Build)  
• Git • Maven

## EXPERIENCE

### HSBC TECHNOLOGY INDIA | SOFTWARE ENGINEER I

Jul 2024 -- Present | Pune

- Architected and deployed an enterprise-wide **RAG pipeline** serving **50+** onboarded applications on the **iQube** platform. Improved technical documentation retrieval accuracy by **40%** and reduced developer support queries by **30%**.
- Engineered an event-driven **Jenkins CI/CD** workflow integrated with LLMs to auto-generate and index documentation on repository commits, eliminating manual doc-sync overhead and ensuring 100% synchronization.
- Optimized the RAG search engine using advanced semantic chunking, distributed vector embeddings, and cross-encoder re-ranking, boosting p95 retrieval latency by **45%** for domain-specific queries.
- Led the design and development of the centralized **Auth module** for iQube. Implemented fine-grained, Entitlement and Role-Based Access Control (RBAC), securing **200+** API endpoints and handling **15,000+ daily requests** with sub-50ms latency.
- Redesigned the data-access layer for a core financial dataset of **1 Million** records. Replaced offset-based pagination with composite-indexed, cursor-based filtering and Redis caching, slashing query execution time from **12s** to **<800ms**.
- Resolved critical N+1 query bottlenecks in a key reconciliation flow, optimizing the execution plan to bring end-to-end processing latency down from **10s** to **1.5s**, validated via extensive SRE load testing.
- Spearheaded system integration for **15+** cross-functional teams across Gateway, Notification, and Workflow (Flowable BPMN) modules. Standardized REST contracts and Kafka event schemas, reducing average onboarding time from **8 weeks** to under **5 days**.

## PROJECTS

### RA CONNECT | SAAS PLATFORM | LIVE

- Designed and built a multi-tenant SaaS platform for quantitative traders, distributing curated market insights and trade alerts to **100+** active subscribers with strict tenant-level data isolation.
- Implemented a scalable microservices architecture deployed on **GCP Cloud Run**, utilizing Cloud SQL and role-based access control. Scale-to-zero autoscaling reduced operational costs by **85%** while maintaining 99.9% uptime.

### MLDRILLS | ML PRACTICE PLATFORM | LIVE | GITHUB

- Built and deployed an interactive ML education platform serving **30+** active users, providing instant feedback on hands-on Python/PyTorch code submissions.
- Engineered a secure, isolated in-browser execution environment for AI/NLP tasks (Transformers, Diffusion models), significantly reducing server compute overhead and ensuring instantaneous user feedback.

### VEIL | REDUCE PHONE USAGE

- Developed a cross-platform mobile application using **Flutter** to intercept and enforce screen-time limits in real-time, leveraging native platform channels for background processing.
- Optimized the system-wide overlay for minimal battery drain and zero frame-drops, ensuring smooth performance when blocking restricted apps.